LECTURE 8

SIMPLE MACHINES

SIMPLE MACHINES

- Machines make work easier.
- They normally transform input forces into output forces in a more convenient form for a required task.
- Simple machines in mechanics fall into six broad categories:
  - Lever
  - Inclined plane
  - Wedge
  - Screw
  - Wheel and axle
  - Pulley
- Compound machines comprise two or more simple machines that work together to achieve a required task.

SIMPLE MACHINES

Some Terms and Definitions

- Effort – force applied to do work at on some part of the machine.
- Load – external force that is overcome by the effort in doing work.
- Effort and Load Distances – distances effort and load must move through in order for work to be done.
- Mechanical Advantage – ratio of load to effort.
- Velocity Ratio – ratio of effort distance to load distance.
- For an ideal machine, work done by the effort is equal to work that should be done on the load.

$$P \times \text{distance through which } P \text{ moves} = W \times \text{distance through which } W \text{ moves}$$

$$\therefore \frac{W}{P} = \frac{\text{distance through which } P \text{ moves}}{\text{distance through which } W \text{ moves}} = Velocity\ Ratio$$

$$M.A. = V.R.$$

- Ideal machines don't exist in real life.

SIMPLE MACHINES
Some Terms and Definitions

- Efficiency of the machine – ratio of useful work done to work supplied.

$$Efficiency = \frac{Useful\ work\ done\ by\ the\ machine}{work\ input\ to\ the\ machine}$$

If x and y are the distances moved through by the Effort, (P) and the Load, (W) respectively, then

$$Efficiency = \frac{Wy}{Px}$$

$$= \frac{W/P}{x/y} = \frac{Mechanical\ Advantage}{Velocity\ Ratio}$$

SIMPLE MACHINES
Overhauling in Machines

- Overhauling in simple machines refers to a scenario in which the when the effort on the machine is removed, it turns to run back with the load now acting as the effort.
- Machines must be *self-locking* or *self-sustaining* in order for this not to happen
- Machines overhaul only if their efficiency is greater than 50%.
- Proof:

For a machine running a direction assumed to be the foward direction,

$$E_{in} = E_{out} + E_{loss}$$

$$Efficiency = \frac{E_{out}}{E_{in}} = \frac{E_{out}}{E_{out} + E_{loss}} = \frac{1}{1 + \left(\frac{E_{loss}}{E_{out}} \right)}$$

SIMPLE MACHINES
Overhauling in Machines

- If this machine runs backward now, with the effort removed,

Work done in lowering the load $= E_{out}$

Energy available for this $= E_{out} - E_{loss}$

If $E_{in} = E_{out} > E_{loss}$

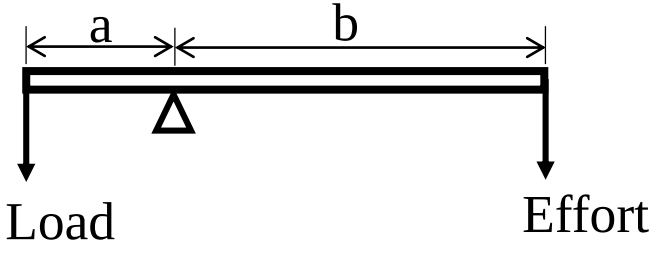
$$Efficiency = \frac{E_{out}}{E_{in}} = \frac{E_{out}}{E_{out} + E_{loss}} = \frac{1}{1 + \left(\frac{E_{loss}}{E_{out}} \right)} => 50\%$$

- Else, it means the losses in the system are so great the amount of energy the load supplies to the machine when it is running backwards, is too small and can't overcome them. Hence, the machine can't overhaul, so it will ***self-lock.***

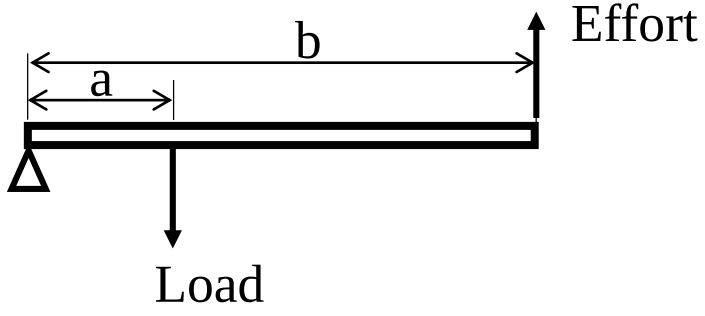
SIMPLE MACHINES

The Lever

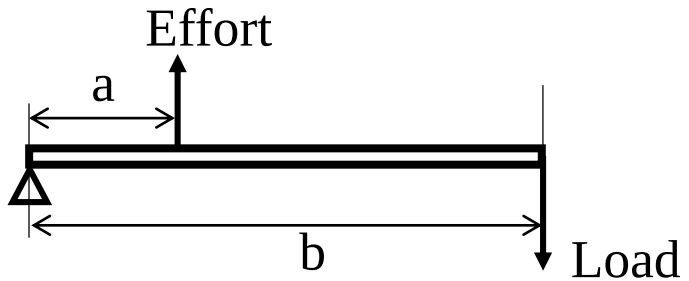
- A rigid member that is capable or rotating about a fulcrum such that the effort at one point can overcome a load at another point.
- Classified according to positions of effort and load
- The relationship between load and effort is given by $Wa = Pb$



1st Class



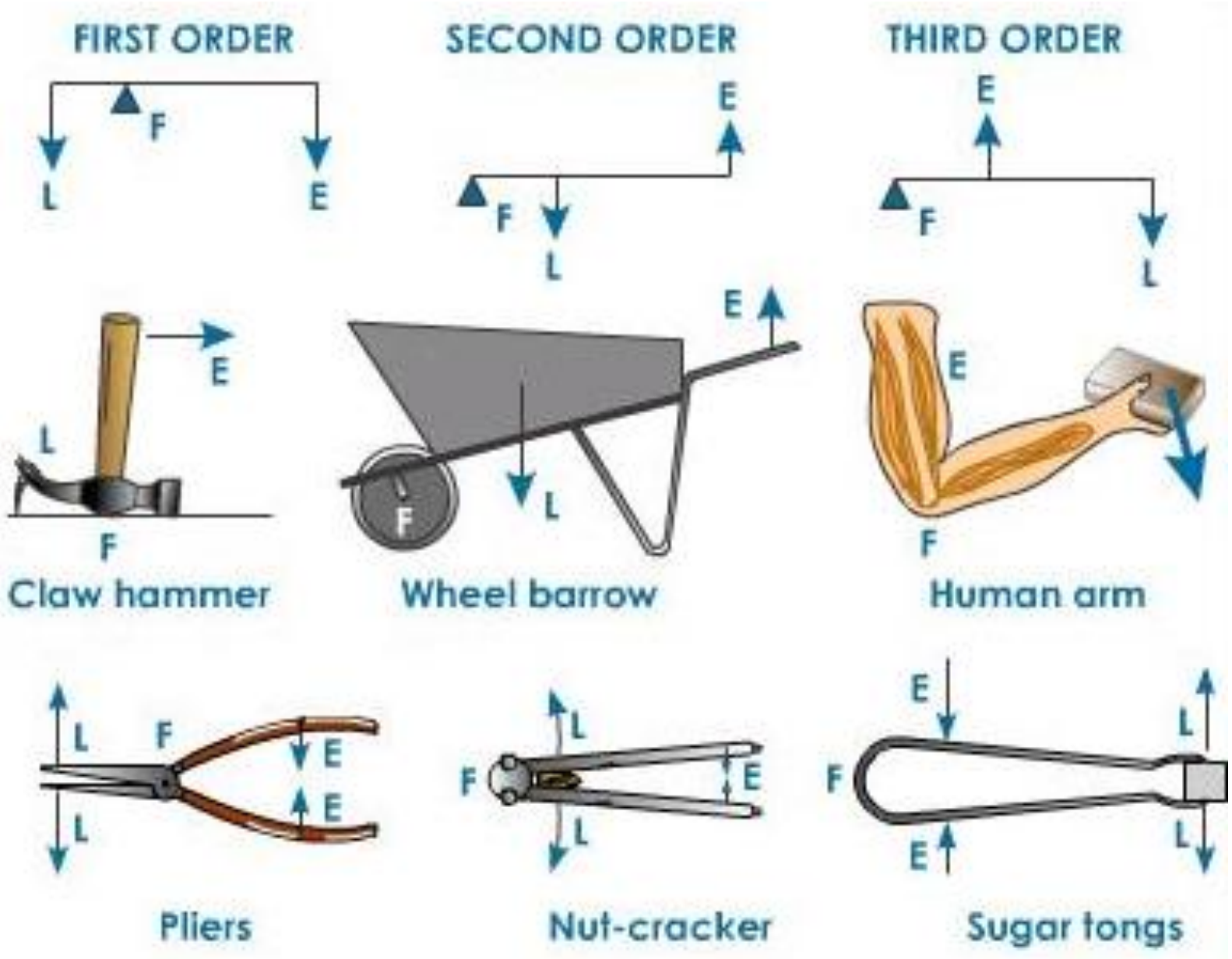
2nd Class



3rd Class

SIMPLE MACHINES

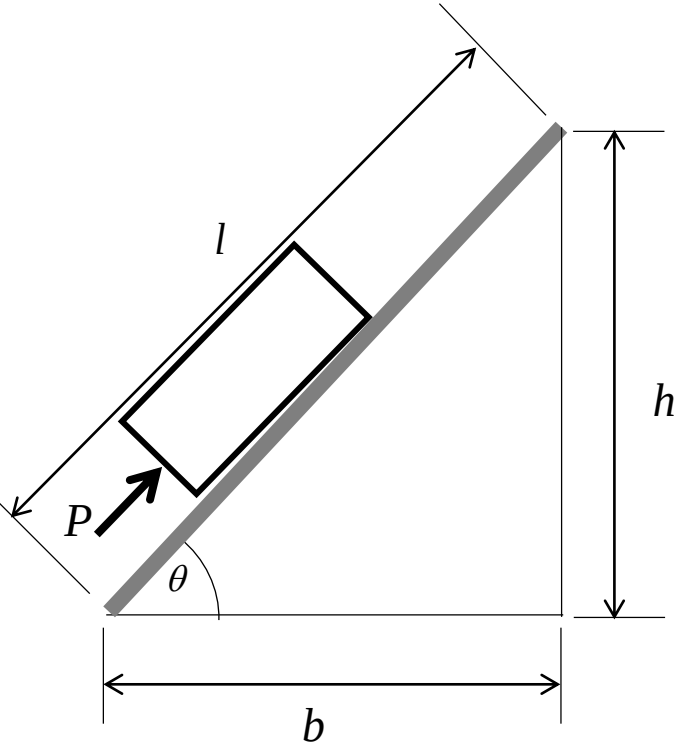
The Lever

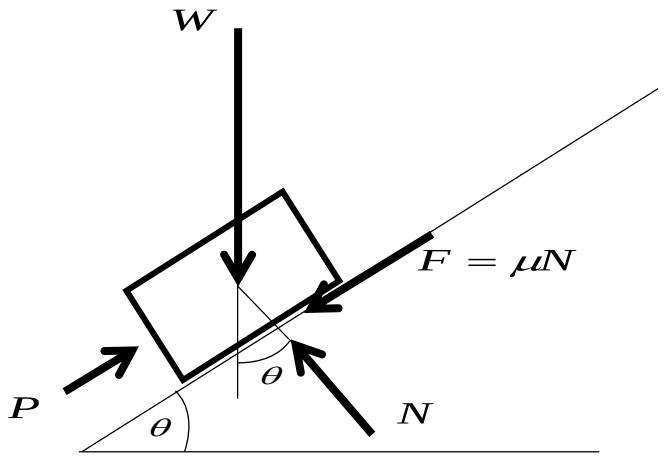


SIMPLE MACHINES

The Inclined Plane

- Normally used to move heavy bodies through a vertical distance by moving it up an inclined plane.





FBD

Summing forces,

$$+\nearrow \sum \mathrm{F}_{\text{parallel}} = 0;\ P - W\sin\theta - F = 0$$

$$+\nwarrow \sum \mathrm{F}_{\text{perpendicular}} = 0;\ W\cos\theta - N = 0$$

$$P = W(\sin\theta + \mu\cos\theta)$$

$$M.A. = \frac{W}{P} = \frac{1}{(\sin\theta + \mu\cos\theta)}$$

SIMPLE MACHINES

The Inclined Plane

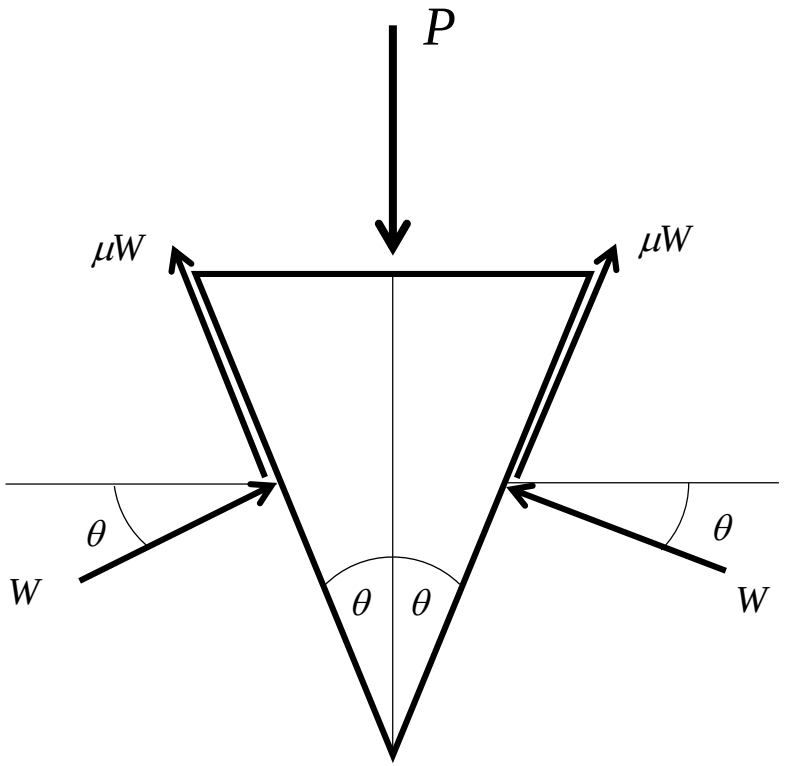
- Example
- A 100 kg crate is being loaded onto the bed of a pickup truck using a ramp. The floor of the truck's bed is 1.2 m from the ground. If the crate is being pushed with a force of 30N, determine the angle of inclination, as well as the length of the ramp. Take the co-efficient of static friction between the crate material and ramp material to be 2.5.

SIMPLE MACHINES

The Wedge

- This is basically two inclined planes set together.





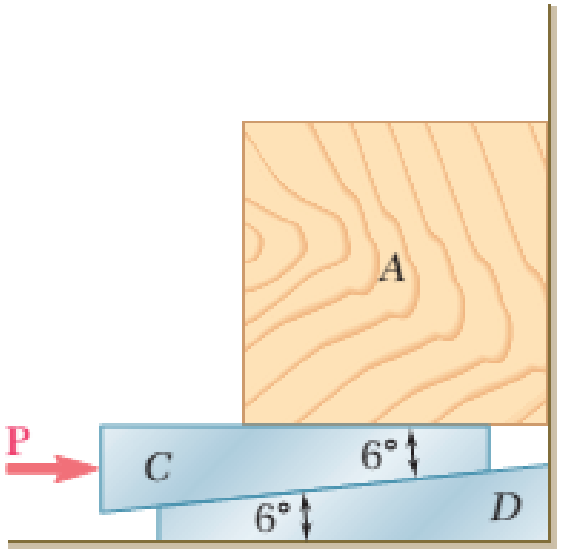
$$P = 2W(\sin\theta + \mu\cos\theta)$$

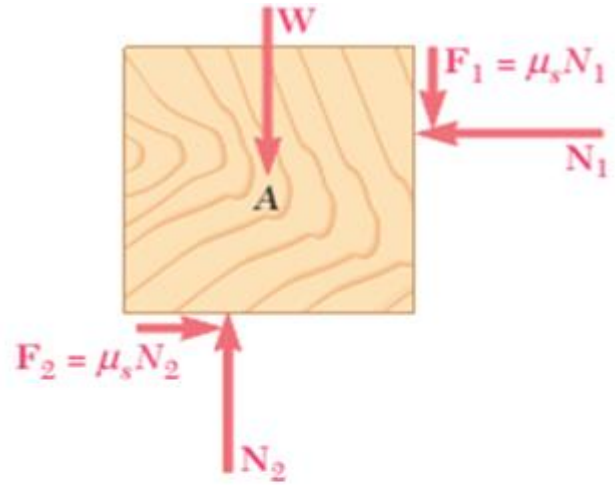
$$M.A. = \frac{W}{P} = \frac{1}{2(\sin\theta + \mu\cos\theta)}$$

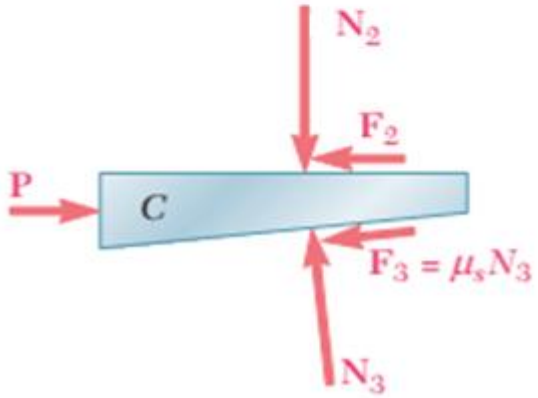
SIMPLE MACHINES

The Wedge

- Wedges are also used to raise loads.







SIMPLE MACHINES

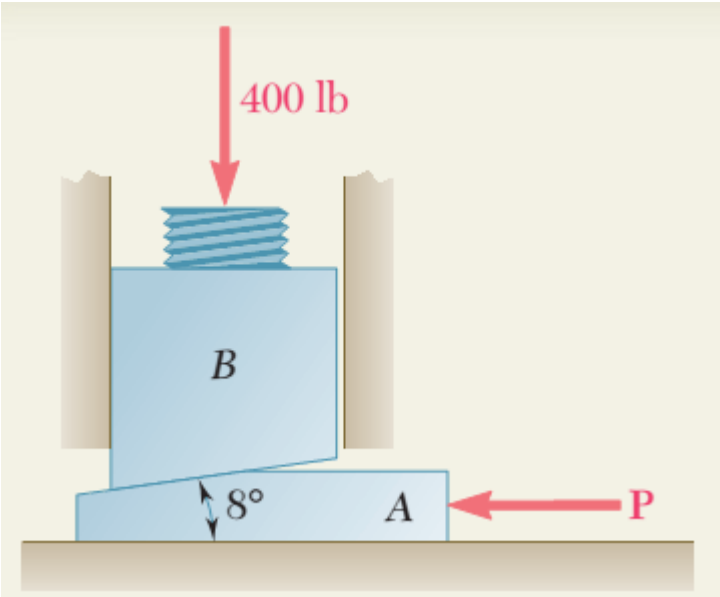
The Wedge

➢Example

The position of the machine block B is adjusted by moving the wedge A. Knowing that the coefficient of static friction is 0.35 between all surfaces of contact, determine the force P required

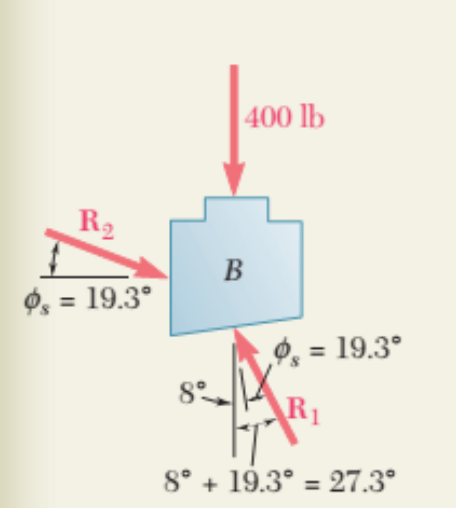
(a) to raise block B,

(b) to lower block B.



SIMPLE MACHINES
The Wedge

➢Example – Solution



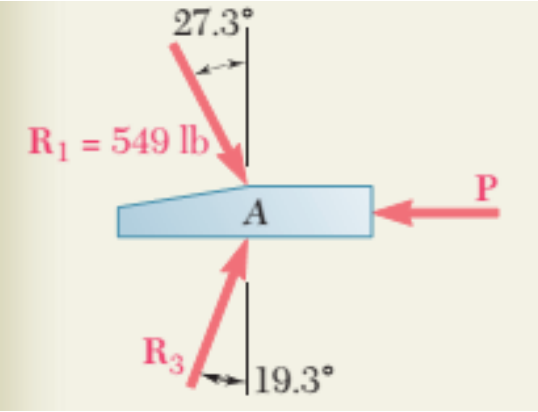
Summing forces,

$$+\rightarrow \sum F_x = 0; -R_1 \sin 27.3^o + R_2 \cos 19.3^o = 0 \qquad ---(1)$$

$$+\uparrow \sum F_y = 0; R_1 \cos 27.3^o - R_2 \sin 19.3^o - 400\,\text{lb} = 0 \qquad ---(2)$$

Solving simultaneously,

$$R_1 = 549\,\text{lb}$$



Summing forces,

$$+\rightarrow \sum F_x = 0; R_1 \sin 27.3^o + R_3 \sin 19.3^o - P = 0 \qquad ---(1)$$

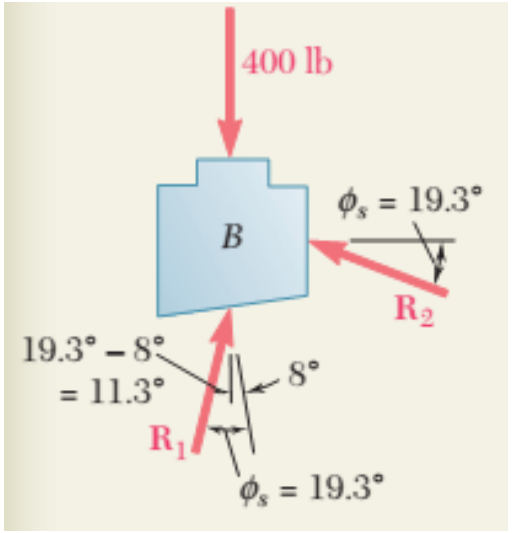
$$+\uparrow \sum F_y = 0; R_3 \cos 19.3^o - R_1 \cos 27.3^o = 0 \qquad ---(2)$$

Solving simultaneously,

$$P = 423\,\text{lb}$$

SIMPLE MACHINES
The Wedge

➢Example – Solution



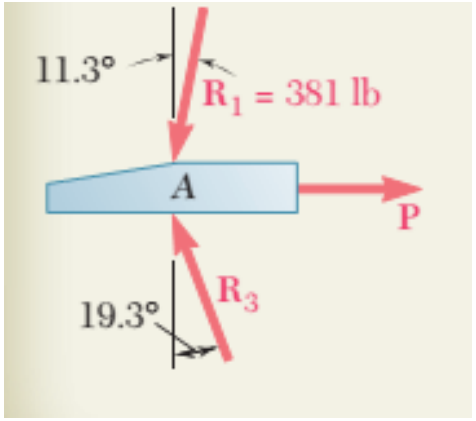
Summing forces,

$$+\rightarrow \sum F_x = 0;\ R_1 \sin 11.3.^o - R_2 \cos 19.3^o = 0 \qquad ---(1)$$

$$+\uparrow \sum F_y = 0;\ R_1 \cos 11.3^o + R_2 \sin 19.3^o - 400\,\text{lb} = 0 \qquad ---(2)$$

Solving simultaneously,

$$R_1 = 381\,\text{lb}$$



Summing forces,

$$+\rightarrow \sum F_x = 0;\ -R_1 \sin 11.3^o - R_3 \sin 19.3^o + P = 0 \qquad ---(1)$$

$$+\uparrow \sum F_y = 0;\ R_3 \cos 19.3^o - R_1 \cos 11.3^o = 0 \qquad ---(2)$$

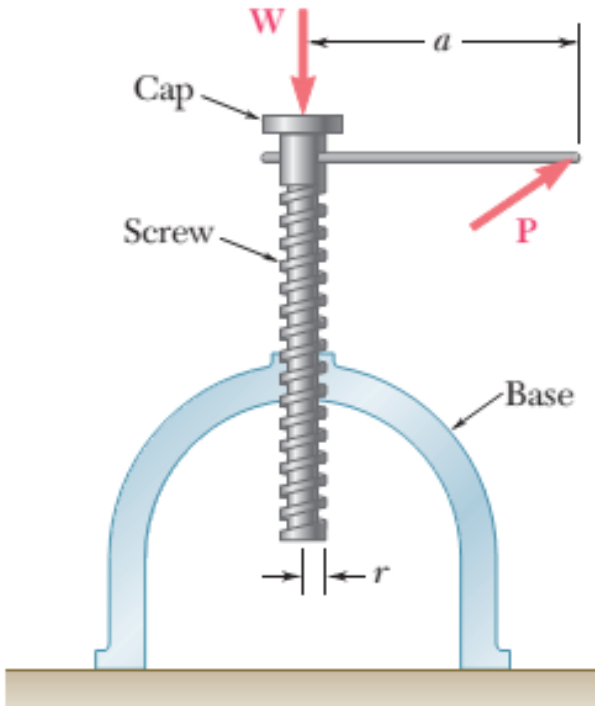
Solving simultaneously,

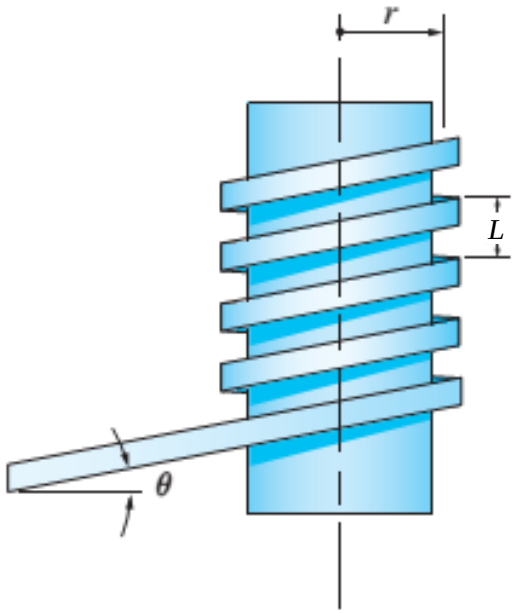
$$P = 206\,\text{lb}$$

SIMPLE MACHINES

The Screw

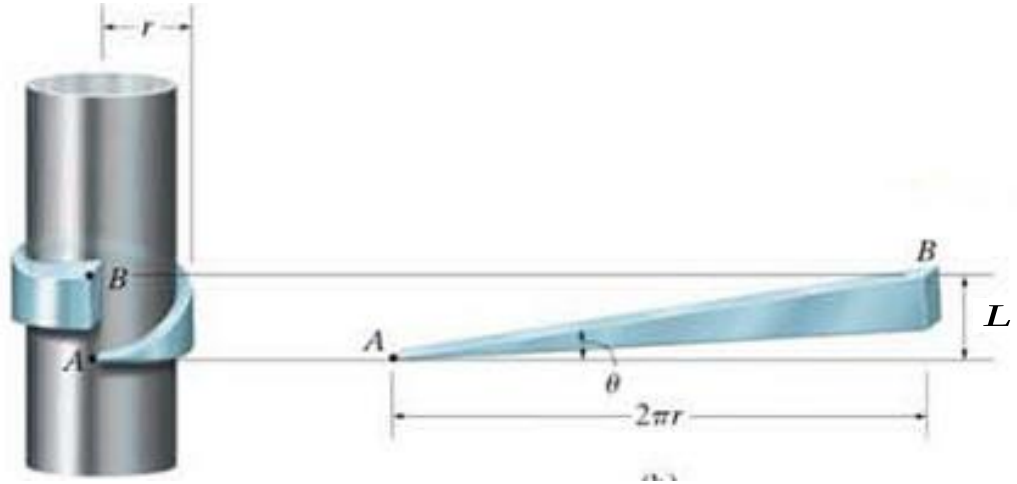
- Normally used as fasteners. But square-threaded screws are also often used for transmitting power.
- Can be considered as an inclined plane wrapped around a shaft.
- Often used in jacks, presses, vices, clamps, etc.



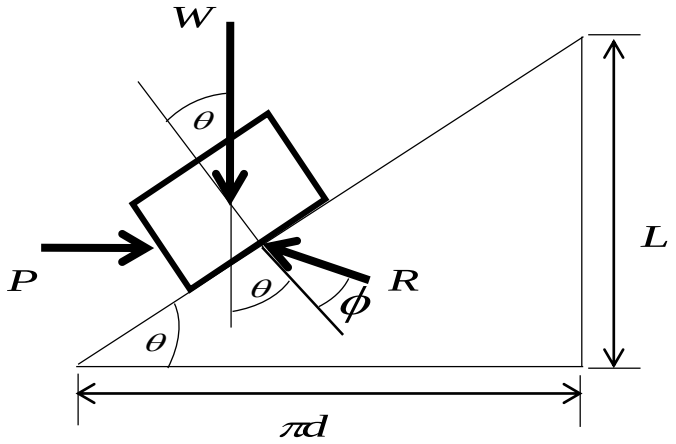


SIMPLE MACHINES

The Screw



- Screwing a nut against the load (moving the load up)



FBD

For equilibriu m,

$$+\rightarrow \sum F_x = 0;\ P - R\sin(\phi+\theta) = 0 \qquad ---(1)$$

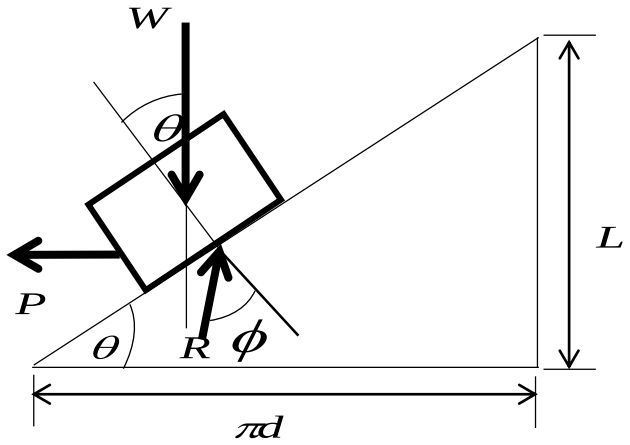
$$+\uparrow \sum F_y = 0;\ -W + R\cos(\phi+\theta) = 0 \qquad ---(2)$$

Eliminatin g R and simplifyin g, we obtain

$$P = W\tan(\phi+\theta)$$

SIMPLE MACHINES
The Screw

- For a load to move down



FBD

For equilibriu m,

$$+\rightarrow \sum F_x = 0; -P + R\sin(\phi - \theta) = 0 \quad ---(1)$$

$$+\uparrow \sum F_y = 0; -W + R\cos(\phi - \theta) = 0 \quad ---(2)$$

Eliminatin g R and simplifyin g, we obtain

$$P = W\tan(\phi - \theta)$$

- Load won't move down until a force, P is applied as shown above. This is the situation in most practical scenarios, where screws **self-locking**. The loads self-locking screws support don't move down until a force is applied to move them down. Screws are also self locking when $\emptyset = \Theta$.
- If , $\emptyset < \Theta$, the load moves down on it's own, **without** the application of P as shown above. The screw is then not **self-locking**.

SIMPLE MACHINES

The Screw

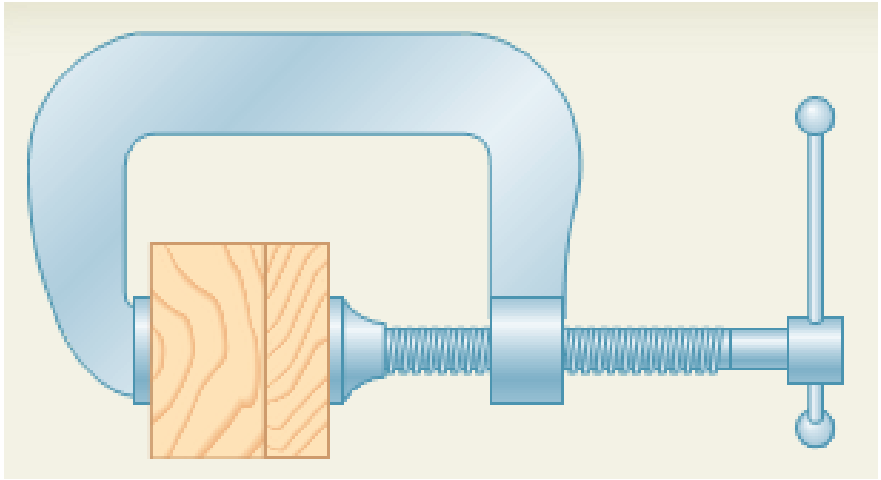
Example

A clamp is used to hold two pieces of wood together as shown. The clamp has a double square thread of mean diameter equal to 10 mm with a pitch of 2 mm. The coefficient of friction between threads is $\mu_s = 0.30$.

If a maximum couple of 40 Nm is applied in tightening the clamp, determine

(*a*) the force exerted on the pieces of wood,

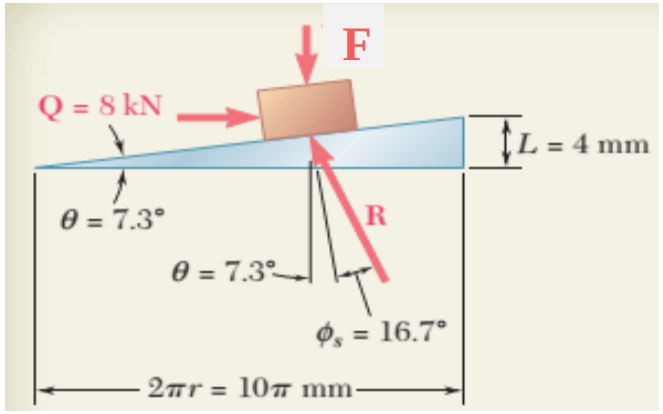
(*b*) the couple required to loosen the clamp.



SIMPLE MACHINES
The Screw

Example – Solution

Force exerted on wood in tightening the clamp



Summing forces,

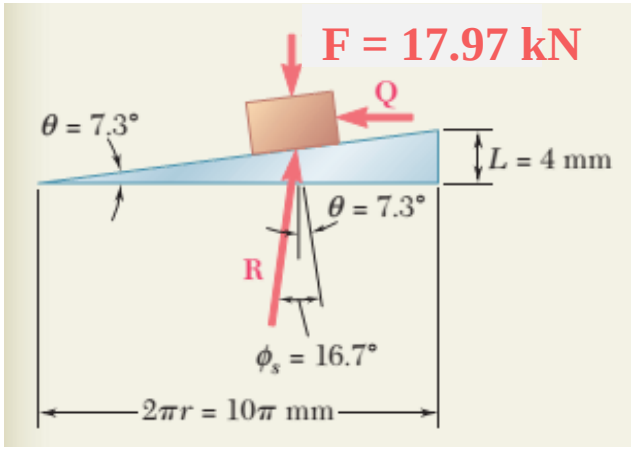
$$+\rightarrow \sum F_x = 0; 8\text{ kN} - R\sin\left(16.7^o + 7.3^o\right) = 0 \qquad ---(1)$$

$$+\uparrow \sum F_y = 0; -F + R\cos\left(16.7^o + 7.3^o\right) = 0 \qquad ---(2)$$

Solving simultaneously,

$F = 17.96$ kN

To loosen the clamp,



Summing forces,

$$+\rightarrow \sum F_x = 0; Q - R\sin 9.4^o = 0 \qquad ---(1)$$

$$+\uparrow \sum F_y = 0; -17.97\text{ kN} + R\cos 9.4^o = 0 \qquad ---(2)$$

Solving simultaneously,

$Q = 2.97$ kN

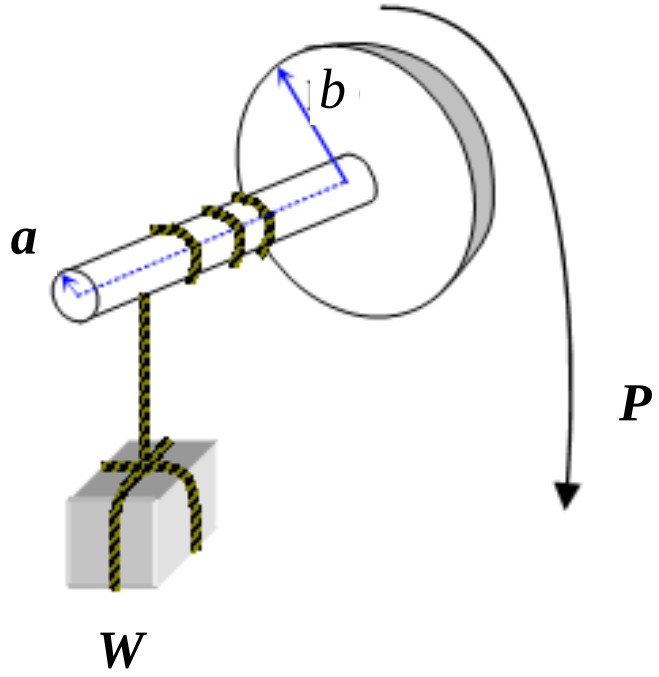
$$\text{Couple Required} = Q \times r = 2.97\text{ kN} \times 0.005\text{ m} = 14.85\text{ Nm}$$

SIMPLE MACHINES

The Wheel and Axle

- Comprises a wheel attached to a shaft (axle) of smaller diameter. The axle turns the wheel.
- Effort is applied to the wheel via a rope for instance.
- Load is attached to axle.

$$Wa = Pb$$



- There is also the differential wheel and axle. This helps achieve a bigger MA

SIMPLE MACHINES

STUDENTS SHOULD READ ON WHEEL AND AXLES AND PULLEYS.

ASSIGNMENT III